STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 7: Gravitation

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. Kepler’s law of equal areas?
2. Universal gravitation law?
3. SI unit of G?
4. g at surface of a spherical planet?
5. Gravitational potential energy zero at infinity?
6. Does escape speed depend on launched mass?
7. What supplies centripetal force to a satellite?
8. Kepler’s third law for fixed central mass?

Section B 2-mark questions - 6 x 2 = 12 marks

9. Force between 2 kg and 3 kg 1 m apart; G=6.67 x 10-.
10. Weight of 50 kg for g=9.8?
11. If orbit radius doubles, circular-speed factor?
12. At altitude h=R, ratio g_h/g?
13. Approximate potential-energy change for small height h?
14. If orbital radius becomes 4r, period factor?

Section C 3-mark questions - 4 x 3 = 12 marks

15. Find g at height h=R/2 relative to surface g.
16. Escape speed for g=5 m/s^2 and R=2 x 10 m?
17. GM=4 x 10, r=7 x 10 m: orbital speed?
18. Weight is 600 N at surface. Weight at h=R?

Section D 4-mark questions - 2 x 4 = 8 marks

19. Derive circular orbital speed and period.
20. kg satellite at r=7 x 10 m, GM=4 x 10. Find speed and energy.